Emotional Models: Supervised vs. Zero-Shot Classification

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1 Introduction

This project presents the implementation and evaluation of two distinct approaches to emotion classification using Transformer-based models applied to the *Emotion* dataset.
The main objective of this project is to compare the performance of two different methodologies for emotion classification. The first approach follows a supervised learning paradigm, where pretrained Transformer models are used as frozen feature extractors to generate sentence embeddings via models from the `sentence-transformers` library. The second approach adopts a zero-shot learning strategy based on Natural Language Inference (NLI) models, where emotion labels are expressed as natural language hypotheses and classification is performed without task-specific fine-tuning. By evaluating both supervised and zero-shot methods on different pretrained models from the HuggingFace Hub using the same dataset, this project aims to analyze their respective strengths and limitations, as well as characteristic error patterns.

2 Description of the problem

The `dair-ai/emotion` dataset consists of text samples, primarily collected from Twitter, labeled with one of six emotions: *joy, sadness, anger, fear, love* and *surprise.* The samples are distributed according to the following table:

Table 1. Class distribution of the `dair-ai/emotion` dataset.

Emotion	Train	Validation	Test	Total (%)
Joy	5,362	704	695	33.8%
Sadness	4,666	550	581	29.0%
Anger	2,159	275	275	13.5%
Fear	1,937	212	224	11.9%
Love	1,304	178	159	8.2%
Surprise	572	81	66	3.6%
Total	**16,000**	**2,000**	**2,000**	**100.0%**

By looking at table 1, it's clear that the dataset is imbalanced with the *joy* and *sadness* classes being significantly more represented. Together, they account for over 60% of the data. In contrast, the *surprise* class is the least represented by a considerable margin, 3.6%. This imbalance could lead to biased model performance, favoring the majority classes, and makes it clear that F1 score is a more appropriate metric than accuracy for evaluating model performance on this dataset. Another issue present is the mislabeling of some samples. After analysing the dataset, we found samples labeled with the wrong emotions. For example, the sentence "I didn't feel humiliated" was labeled as *sadness*. The sentence explicitely denies the negative emotion, yet it was labeled with it, presumably due to the presence of the word "humiliated". This kind of mislabeling will inevitably cap the performance of any model trained on this dataset, as it introduces noise and confusion during the learning process. Another possible cause is the fact that dataset relies on hastags to label the samples, and their use in sarcastic or ironic contexts is a major source of error.

3 Approach of Supervised Classification

This approach follows a supervised learning paradigm in which pretrained Transformer models are used as fixed feature extractors, and a lightweight neural classifier is trained on top of the resulting sentence embeddings.

3.1 Embedding Generation

The sentence-transformer models are used to encode each sentence into a dense, fixed-length embedding vector. The embedding process is performed in batches to improve efficiency and introduces beneficial stochasticity into the optimization process. The Transformer remains frozen during this stage, meaning that no gradient updates are applied to its parameters. These embeddings serve as high-level semantic features that capture syntactic and contextual information from the input text. The corresponding emotion labels are retained as integer class indices, resulting in pairs of the form *(embedding, label)*, which are later wrapped into `DataLoader` objects for mini-batch training and evaluation. On top of the generated embeddings, we trained and evaluated two light classification heads, a Multi-Layer Perceptron and Logistic Regression.

3.2 Classifier Architecture (ANN)

Its architecture is intentionally lightweight, as the main representational power is delegated to the pretrained Transformer embeddings. The architecture consists of:

– **Input Layer:** Matches the dimension of the embeddings.

- **Hidden Layer:** A fully connected layer with 256 neurons, followed by a Rectified Linear Unit (ReLU) activation function, introducing non-linearity into the model.
- **Regularization:** A Dropout layer, implemented in the training, with a probability of $p = 0.1$ to prevent overfitting.
- **Output Layer:** A fully connected layer with six neurons, one for each emotion class, producing unnormalized logits.

Class predictions are obtained by applying the `argmax` operation over the output logits, selecting the emotion class with the highest predicted score.

3.3 Classifier Architecture (Logistic Regression)

As an alternative to the neural classifier, a multinomial Logistic Regression (`LogReg`) model is employed as a lightweight linear classification head on top of the frozen Transformer embeddings. This approach allows us to assess how well linearly separable the embedding space is for the emotion classification task. The Logistic Regression classifier is implemented using the `scikit-learn` library and trained directly on the precomputed sentence embeddings, without any additional feature transformation or hidden layers. Its architecture can be summarized as follows:

- **Input Layer:** A fixed-size embedding vector whose dimensionality matches the output of the pretrained Transformer model.
- **Linear Decision Function:** A single linear transformation that maps the input embeddings to six output scores, one for each emotion class.
- **Multinomial Softmax:** The class probabilities are obtained via the multinomial logistic `softmax` formulation.

4 Approach of Zero-Shot Classification

The zero-shot classification approach leverages pre-trained Natural Language Inference (NLI) models to classify text without specific task-related training data. NLI models are initially trained to determine the relationship between a premise and a hypothesis. For this purpose, the template `"This text expresses ."` is used (`NLI.py`).

4.1 Architecture

1. **Batch Inference:** The test texts are processed in batches for computational efficiency.
2. **Prediction:** For each text, the model compares the text (premise) against a hypothesis for every candidate label.
3. **Label Selection:** The label whose hypothesis yields the highest entailment score is selected as the predicted class for the input text.

5 Experimental Setup

5.1 ANN

The classifier is trained using the `Cross-Entropy` loss function, which is standard for multi-class classification tasks. To prevent overfitting and ensure robust model selection, **early stopping** is employed based on the macro-averaged F1 score computed on the validation set. Training is halted if no improvement in validation F1 is observed for a fixed number of consecutive epochs, and the best-performing model state is restored before final evaluation. The ANN utilized in supervised training was configured with the hyperparameters listed in Table 2. The `all-MiniLM-L6-v2`, `paraphrase-mpnet-base-v2` and `all-mpnet-base-v2` models from the `sentence-transformers` library were employed to generate sentence embeddings. Their configurations, including embedding dimensions and batch sizes, are summarized in Table 3. Experiments were conducted using five different seeds to account for variability in training outcomes.

Table 2. Hyperparameters for supervised training

Hyperparameter	**Value**
Batch Size	64
Learning Rate	1×10^{-3}
Epochs	25 with Early Stopping
Loss Function	Cross-Entropy Loss
Optimizer	Adam
Seeds	[2025, 2029]

Table 3. Supervised Models and their configurations.

Model	**Embedding Dimension**
all-MiniLM-L6-v2	384
paraphrase-mpnet-base-v2	768
all-mpnet-base-v2	768

The selected model is evaluated on the held-out test set using accuracy, macro-averaged F1 score, and confusion matrices.

6 Logistic Regression

The classifier is trained using the `lbfgs` optimizer with a maximum of 200 iterations and the `multinomial` setting, which jointly optimizes all class decision boundaries. Unlike the ANN-based classifier, the Logistic Regression head does not require iterative epoch-based training or early stopping. As a result, the model is trained once per embedding configuration using the full training set. Although experiments are repeated across multiple random seeds for consistency with the ANN evaluation protocol, reported metrics for this classifier exhibit zero standard deviation across runs. This behavior is expected, as Logistic Regression

is a deterministic linear model and when trained on identical embeddings and labels, it yields the same learned parameters and predictions regardless of the random seed.
After training, the Logistic Regression classifier is evaluated on the test set using accuracy, macro-averaged F1 score, and confusion matrices.

7 NLI

As described in Section 3.1, the zero-shot classification approach utilizes pre-trained NLI models to perform emotion classification without task-specific training data. The models evaluated in this approach are summarized in Table 4. All models were sourced from the HuggingFace Hub and are tested with a batch size of 32 on a GPU device to optimize inference speed.

Table 4. NLI Models used for Zero-Shot Classification

Model Name	**Size (parameters)**
MoritzLaurer/xtremedistil-l6-h256-zeroshot-v1.1-all-33	$\approx$ 13M
joeddav/xlm-roberta-large-xnli	$\approx$ 600M
MoritzLaurer/bge-m3-zeroshot-v2.0	$\approx$ 600M
MoritzLaurer/deberta-v3-xsmall-zeroshot-v1.1-all-33	$\approx$ 71M
MoritzLaurer/ModernBERT-large-zeroshot-v2.0	$\approx$ 400M
facebook/bart-large-mnli	$\approx$ 400M

Unlike the supervised pipeline, the zero-shot approach is applied directly to the test set without any parameter updates. Performance is evaluated using accuracy, macro-averaged F1 score, and confusion matrices, enabling a fair comparison with the supervised methods and providing insight into overall performance as well as robustness to class imbalance.

8 Results and Discussion

8.1 Supervised Models

The following Table 5 summarizes the performance across the different embedding models and classifier heads used in the supervised approach. Results are reported as mean accuracy and mean macro F1 score over the five different seeds, along with the average number of epochs taken for training before early stopping.

Model	Head	Accuracy	Macro F1	Avg. Epochs
all-MiniLM-L6-v2	ANN	0.7315 ± 0.0061	0.6556 ± 0.0057	20.0
all-MiniLM-L6-v2	LogReg	0.6870 ± 0.0000	0.5928 ± 0.0000	N/A
paraphrase-mpnet-base-v2	ANN	0.7089 ± 0.0037	0.6249 ± 0.0091	12.4
paraphrase-mpnet-base-v2	LogReg	0.6950 ± 0.0000	0.6083 ± 0.0000	N/A
all-mpnet-base-v2	ANN	0.7014 ± 0.0019	0.6177 ± 0.0059	13.0
all-mpnet-base-v2	LogReg	0.6760 ± 0.0000	0.5629 ± 0.0000	N/A

Table 5. Performance Comparison of Embedding Models and Classifiers

The ANN classifier consistently outperforms the Logistic Regression head across all embedding models. Even though this gap is smaller in `paraphrase-mpnet-base-v2` model, it made a noticeable difference in the other two. This indicates that the non-linear decision boundaries learned by the ANN provide a significant advantage over the linear separability assumed by Logistic Regression, and suggests that the embedding spaces are not linearly separable for the emotion classification task. It is also noticeable that apart from the `all-MiniLM-L6-v2` model which on average used all 25 epochs, early stopping occurred relatively quickly on the rest, within the first 15 epochs. This suggests that is might've been beneficial to train the first model for longer, as the scores were still rising

Model Comparison Among the embedding models evaluated, the highest performance scores were achived by `all-MiniLM-L6-v2` in both accuracy and macro F1 score, regardless of the classifier head used. However, the performance differences between the embedding models are not substantial, indicating that all three models provide reasonably effective representations for the emotion classification task. The size of the models did also not show much effect on the performance, as the smallest model (with dimension 384) actually performed the best in comparison to the larger ones (with dimension 768). This could be due to the fact that the smaller model is less prone to overfitting.

8.2 NLI Models

The following table (Table 6) summarizes the performance of the different NLI models used in the zero-shot classification approach. Results are reported in terms of accuracy and macro F1 score on the test set.

Table 6. Zero-Shot Classification Results

Model Name	ACC	Macro F1
MoritzLaurer/xtremedistil-l6-h256-zeroshot-v1.1-all-33	0.6120	0.5148
joeddav/xlm-roberta-large-xnli	0.3505	0.3222
MoritzLaurer/bge-m3-zeroshot-v2.0	0.6960	0.6232
MoritzLaurer/deberta-v3-xsmall-zeroshot-v1.1-all-33	0.7045	0.6306
MoritzLaurer/ModernBERT-large-zeroshot-v2.0	0.7090	0.6304
facebook/bart-large-mnli	0.4660	0.4333

Model Comparison: The results indicate significant variability in performance across the different NLI models. The model that achieved the highest macro F1 score of 0.6304 was `MoritzLaurer/ModernBERT-large-zeroshot-v2.0`, closely followed by `MoritzLaurer/deberta-v3-xsmall-zeroshot-v1.1-all-33` with a macro F1 of 0.6306. The worst performing model achieved a macro F1 score of only 0.3222 and was `joeddav/xlm-roberta-large-xnli`.
Interestingly, model size did not correlate strongly with model performance. For example, `MoritzLaurer/deberta-v3-xsmall-zeroshot-v1.1-all-33`, which is relatively small, with $\approx$ 71M parameters, outperformed the largest models such as `joeddav/xlm-roberta-large-xnli` with $\approx$ 600M parameters. Models with similar sizes also exhibited varying performance, like `facebook/bart-large-mnli` and `MoritzLaurer/ModernBERT-large-zeroshot-v2.0`, both around $\approx$ 400M parameters.
This discrepancy is due to a different reason: what each model was taught during pretraining. Taking `joeddav/xlm-roberta-large-xnli` and `deberta` as examples. On the one hand, `xlm-roberta` was trained in a multilingual setting, which makes it less specialized in English tasks, and on standard NLI datasets, focusing on logic reasoning. On the other hand, `deberta` was trained in a monolingual setting, specifically for zero-shot classification tasks, and its training included emotion detection, sentiment, and topic classification, which is far more relevant to our task. This proved to be a major factor when it came to performance and that, given a different task, the results could be completely different.
This difference could also be attributed to the architecture of each model, as some may already have an outdated architecture that doesn't perform as well as more recent ones. For example, `facebook/bart-large-mnli` is based on the BART architecture, which, while powerful, has been surpassed by newer architectures like DeBERTa and ModernBERT in various NLP tasks.

8.3 Supervised vs Zero-Shot Classification

After analyzing the results from both the supervised and zero-shot classification approaches, several key observations can be made regarding the best performing models and their comparative performance. The supervised ANN classifier, despite relying on a comparatively smaller embedding model (`all-MiniLM-L6-v2`), demonstrates superior overall performance when compared to the zero-shot NLI-based approach.In particular, the supervised method achieves higher scores across the main evaluation metrics:

- **Accuracy:** Supervised ANN (mean = 0.7315) outperforms NLI (0.7045).
- **Macro F1:** Supervised ANN (mean = 0.6556) exceeds NLI (0.6306).

Performance Analysis A class-level analysis of the recall (true positive rate) reveals where each model excels, based on the row-normalized confusion matrices. Recall is derived from the diagonal elements of the row-normalized matrices.

Table 7. Recall Comparison on Test Set (Supervised ANN vs. NLI Zero-Shot)

Emotion Class Index	**Supervised ANN Recall** ($\pm\sigma$)	**NLI Zero-Shot Recall** $\pm\sigma = 0$	**Difference (ANN - NLI)**
Joy (0)	0.797 ± 0.010	0.764	$+0.033$
Sadness (1)	0.801 ± 0.007	0.722	$+\mathbf{0.079}$
Anger (2)	0.484 ± 0.016	0.722	$-\mathbf{0.238}$
Fear (3)	0.678 ± 0.024	0.709	-0.031
Love (4)	0.682 ± 0.017	0.683	-0.001
Surprise (5)	0.403 ± 0.015	0.500	$-\mathbf{0.097}$

The data in Table 7 immediately highlights the discrepancy in performance for two of the most critical classes:

- The **Supervised ANN** exhibits the highest recall for the `Joy` and `Sadness` classes, resulting in a considerable advantage of $+\mathbf{0.079}$ for `Sadness` over the NLI model. This suggests that the supervised training effectively utilizes the embeddings to distinguish these two largest classes.
- The **NLI Zero-Shot** model exhibits a clear and substantial advantage in classifying the `Anger` class, with a difference of $-\mathbf{0.238}$ in recall compared to the Supervised ANN. Furthermore, the NLI model performs significantly better on the least represented `Surprise` class ($-\mathbf{0.097}$), indicating a more robust handling of certain minority classes, likely due to the generalized semantic knowledge of the base `NLI` model.

Overall, the Supervised ANN achieved a higher overall Macro F1 score, primarily driven by its superior recall on the largest classes, `Joy` and `Sadness`. However, the NLI model's strong performance on `Anger` and `Surprise` demonstrates the competitive nature of zero-shot transfer learning, especially where the supervised head struggles to learn complex boundaries.

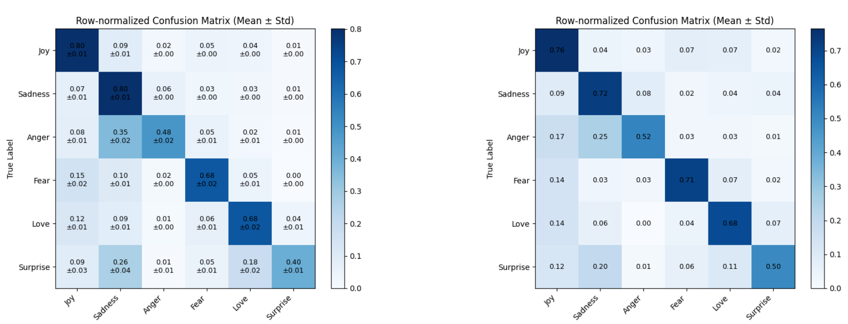

Fig. 1. Covariance percentage heatmaps: Supervised (left) vs. NLI (right).

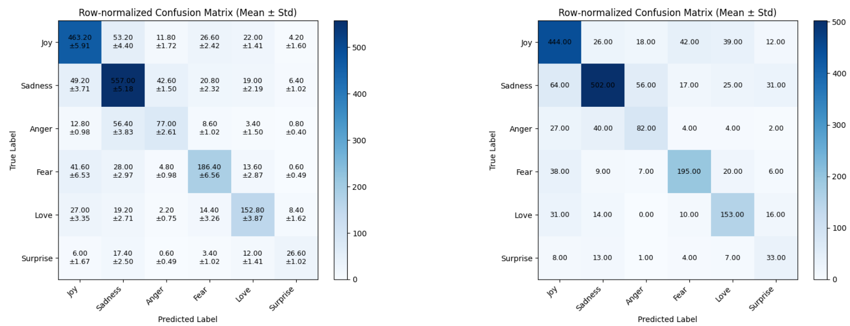

Fig. 2. Covariance values heatmaps: Supervised (left) vs. NLI (right).

The supervised model's most significant error pattern is the misclassification of `Anger` (2), which is directly responsible for the low `Anger` recall.

– **Misclassification of Anger (2):** The single largest source of error in the entire supervised analysis is the misclassification of true `Anger` samples. Approximately 35.5% of texts expressing `Anger` were wrongly predicted as `Sadness`. A further 8.1% were misclassified as `Joy`. This suggests that the ANN classifier, trained on `all-MiniLM-L6-v2` embeddings, maps `Anger` texts into a semantic region that heavily overlaps with `Sadness`.
– **Misclassification of Surprise (5):** The model struggles significantly with the smallest class, `Surprise`, misclassifying 26.4% of its samples as `Sadness` and 18.2% as `Love`.

The NLI model, while robust on `Anger` and `Surprise`, shows greater difficulty distinguishing between other negative emotions.

– **Confusion of Sadness and Love with Joy:** The model misclassifies 9.2% of true `Sadness` samples and 13.8% of true `Love` samples as `Joy`. This indicates a tendency of the NLI model to conflate positive and negative emotions, particularly when the emotional content is subtle or ambiguous. An example of this is the sentence "I feel so blessed and honored that we get to be its parents", as the words "blessed" and "honored" may are ambiguous and could be associated with both emotions.
– **Confusion of Surprise with Sadness:** Similar to the supervised method, the NLI model's primary confusion for `Surprise` is `Sadness`, with 19.7% of `Surprise` samples being misclassified. An example of this are the sentences "i have been feeling overwhelmed with it all and needing to take time out" and "i found myself feeling a bit overwhelmed", both containing the word "overwhelmed", which was likely the reason for the misclassification, as it happens on a few more samples.

9 Conclusion

In this study, we evaluated the performance of supervised learning versus zero-shot classification for emotion detection on the `dair-ai/emotion` dataset. Our experiments demonstrated that the supervised ANN approach, particularly using the `all-MiniLM-L6-v2` embeddings, achieved the highest overall performance with a mean accuracy of 73.15% and a Macro F1 of 0.6556. This superiority was primarily driven by the model's ability to effectively learn the majority classes, `Joy` and `Sadness`. However, the zero-shot NLI approach,specifically utilizing the `MoritzLaurer/deberta-v3-xsmall` model, proved to be a highly competitive alternative, achieving an accuracy of 70.45% and a Macro F1 of 0.6306. The zero-shot model exhibited significantly better generalization on difficult minority classes, outperforming the supervised model by large margins on `Anger` (recall difference of +0.238) and `Surprise`. These results suggest that while supervised fine-tuning still came out on top for overall metric maximization, zero-shot NLI offers a robust and resource-efficient solution for classes where training data is scarce or where the supervised model struggles to separate overlapping semantic boundaries. [1]

[1] Note: To make this project, AI assistance was used to help in the grammar and formulation of some parts of the text, as well as to generate the LaTeXtables and with minor tasks in code implementation.